Input Power

File: input_power.kicad_sch

5V Power Supply

File: 5v_power_supply.kicad_sch

Input Signal

File: input_signal.kicad_sch

Microcontroller

File: microcontroller.kicad_sch

USB UART Bridge

File: USB_UART_Bridge.kicad_sch

RGB LED

File: rgb_led.kicad_sch

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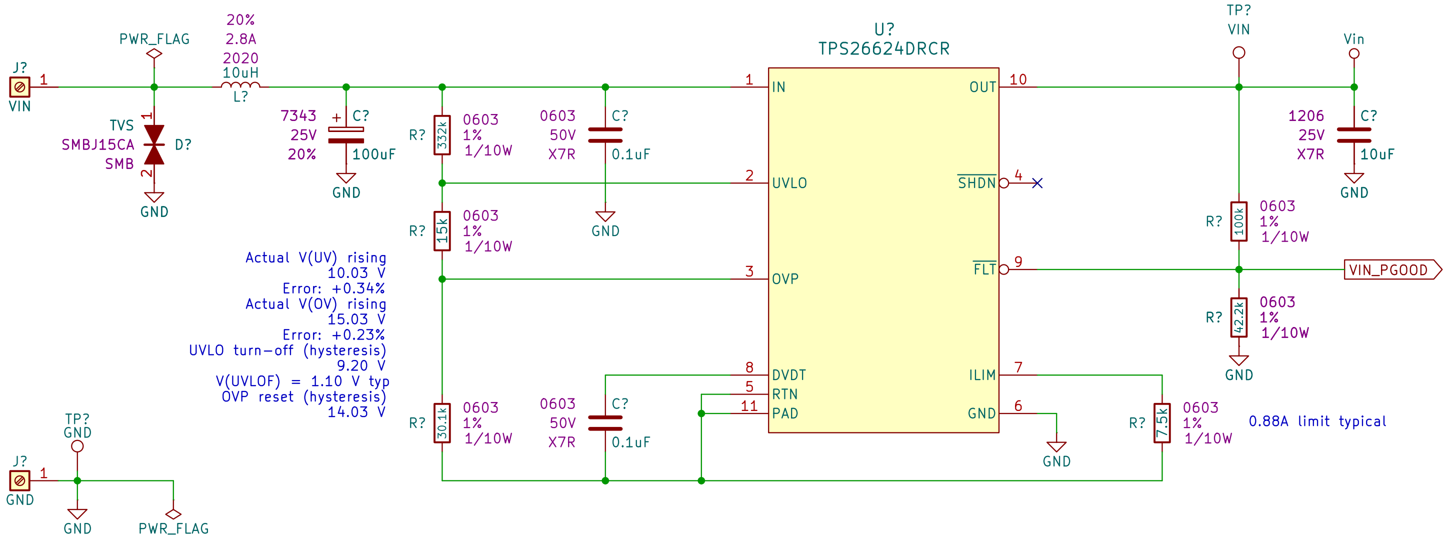
Sheet: /

File: HEI_Shift_Light.kicad_sch

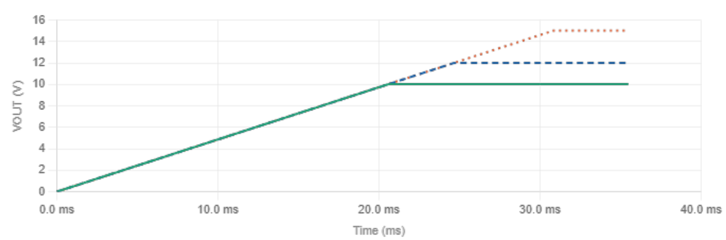
Title: HEI Shift Light

Size: A	Date: 2026-05-29	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 1/7



DVDT cap:
Min corner: I=2.33 µA, GAIN=23.9 V/V (longest ramp)
Typ: I=1.98 µA, GAIN=24.6 V/V
Max corner: I=1.68 µA, GAIN=25.2 V/V (shortest ramp − slowest I, widest gain)

VOUT target Rise time − min Rise time − typ Rise time − max Slew rate (typ)
10.03 V (UVLO threshold) 18.0 ms 20.6 ms 23.7 ms 0.49 V/ms
12 V (Nominal) 21.5 ms 24.6 ms 28.3 ms 0.49 V/ms
15.03 V (OVP threshold) 27.0 ms 30.9 ms 35.5 ms 0.49 V/ms



VIN_PGOOD divider:
VEN at VUV (min VOUT)
2.98 V
Must be ≥ 0.65 V □
VEN at VOV (max VOUT)
4.46 V
□ Within safe range
Divider current at VUV
70.53 µA
Keep ≥ 10× EN leakage
□ V(EN) min = 2.98 V (need ≥ 0.65 V)□ V(EN) max = 4.46 V□ Fault: FLT pulls EN to ~0 V (< 0.3 V)

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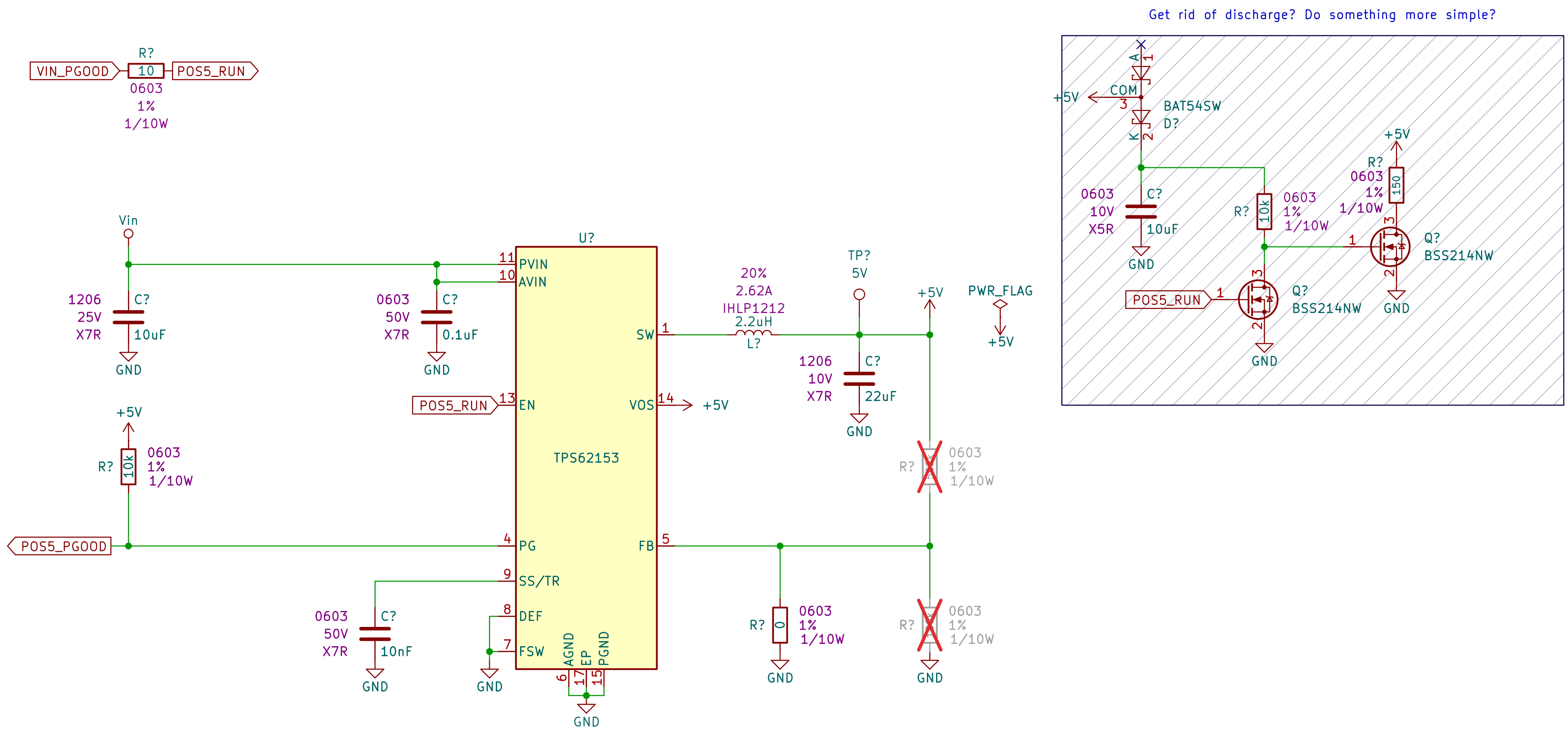
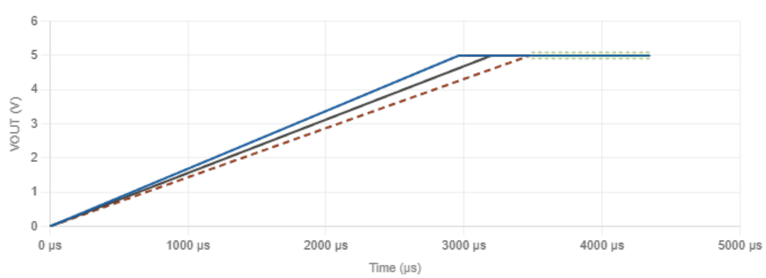


Rise times & settled output range

Min time: I=2.7 µA (fastest) | Typ: I=2.5 µA | Max time: I=2.3 µA (slowest)

Condition	Rise time – min	Rise time – typ	Rise time – max	Settled VOUT
DEF=low (nominal 5.00 V)	2.963 ms	3.200 ms	3.478 ms	4.910 – 5.090 V
DEF=high (+5% = 5.25 V)	2.963 ms	3.200 ms	3.478 ms	5.155 – 5.345 V



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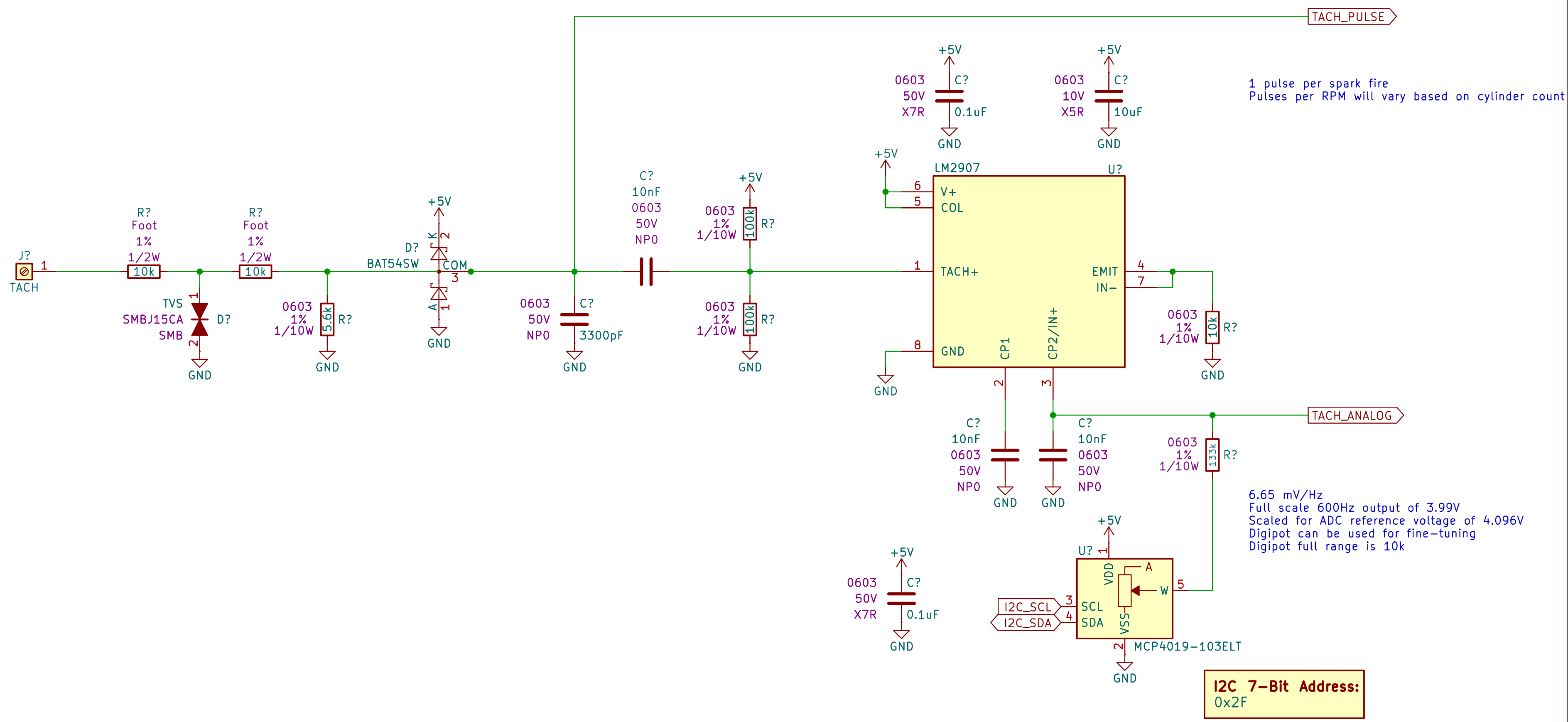
Sheet: /5V Power Supply/
File: 5v_power_supply.kicad_sch

Title: HEI Shift Light

Size: A | Date: 2026-05-29 | Rev: A
KiCad E.D.A. 10.0.3 | Id: 3/7



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Sheet: /Input Signal/

File: input_signal.kicad_sch

Title: HEI Shift Light

Size: A	Date: 2026-05-29	**Rev: A**
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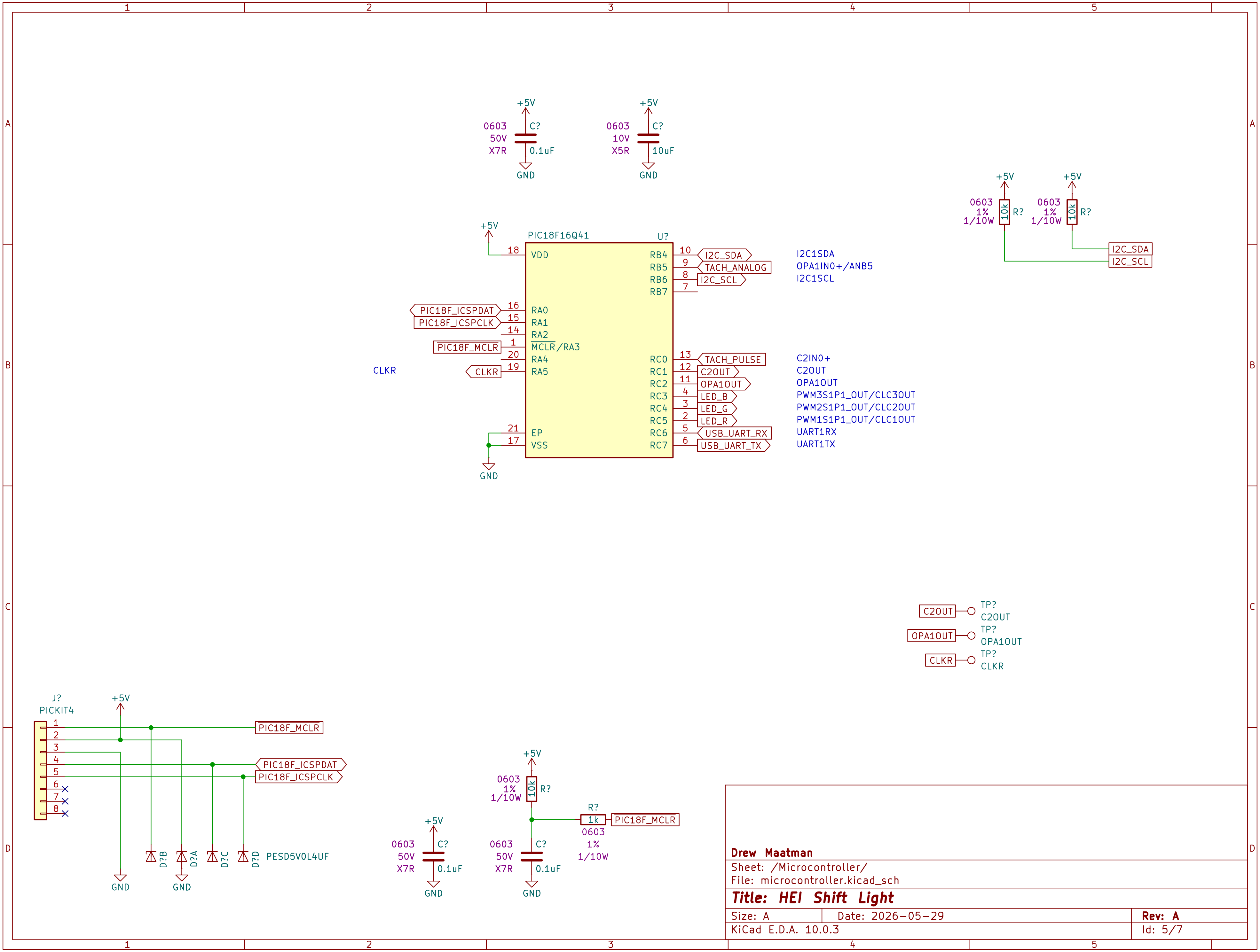
+5V
0603
50V
X7R
C?
0.1uF
GND
+5V
0603
10V
X5R
C?
10uF
GND
+5V
0603
1%
1/10W
10k
R?
+5V
0603
1%
1/10W
10k
R?
I2C_SDA
I2C_SCL
+5V
PIC18F16Q41
U?
18 VDD
RB4 10 I2C_SDA
RB5 9 TACH_ANALOG
RB6 8 I2C_SCL
RB7 7
I2C1SDA
OPA1IN0+/ANB5
I2C1SCL
PIC18F_ICSPDAT 16 RA0
PIC18F_ICSPCLK 15 RA1
14 RA2
PIC18F_MCLR 1 MCLR/RA3
20 RA4
CLKR
CLKR 19 RA5
RC0 13 TACH_PULSE
RC1 12 C2OUT
RC2 11 OPA1OUT
RC3 4 LED_B
RC4 3 LED_G
RC5 2 LED_R
RC6 5 USB_UART_RX
RC7 6 USB_UART_TX
C2IN0+
C2OUT
OPA1OUT
PWM3S1P1_OUT/CLC3OUT
PWM2S1P1_OUT/CLC2OUT
PWM1S1P1_OUT/CLC1OUT
UART1RX
UART1TX
21 EP
17 VSS
GND
C2OUT TP? C2OUT
OPA1OUT TP? OPA1OUT
CLKR TP? CLKR
J?
PICKIT4
+5V
1
2
3
4
5
6
7
8
PIC18F_MCLR
PIC18F_ICSPDAT
PIC18F_ICSPCLK
GND
D?B
GND
D?A
D?C
D?D
PESD5V0L4UF
+5V
0603
50V
X7R
C?
0.1uF
GND
+5V
0603
1%
1/10W
10k
R?
R?
1k
0603
1%
1/10W
PIC18F_MCLR
0603
50V
X7R
C?
0.1uF
GND
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Sheet: /Microcontroller/
File: microcontroller.kicad_sch
Title: HEI Shift Light
Size: A
Date: 2026-05-29
Rev: A
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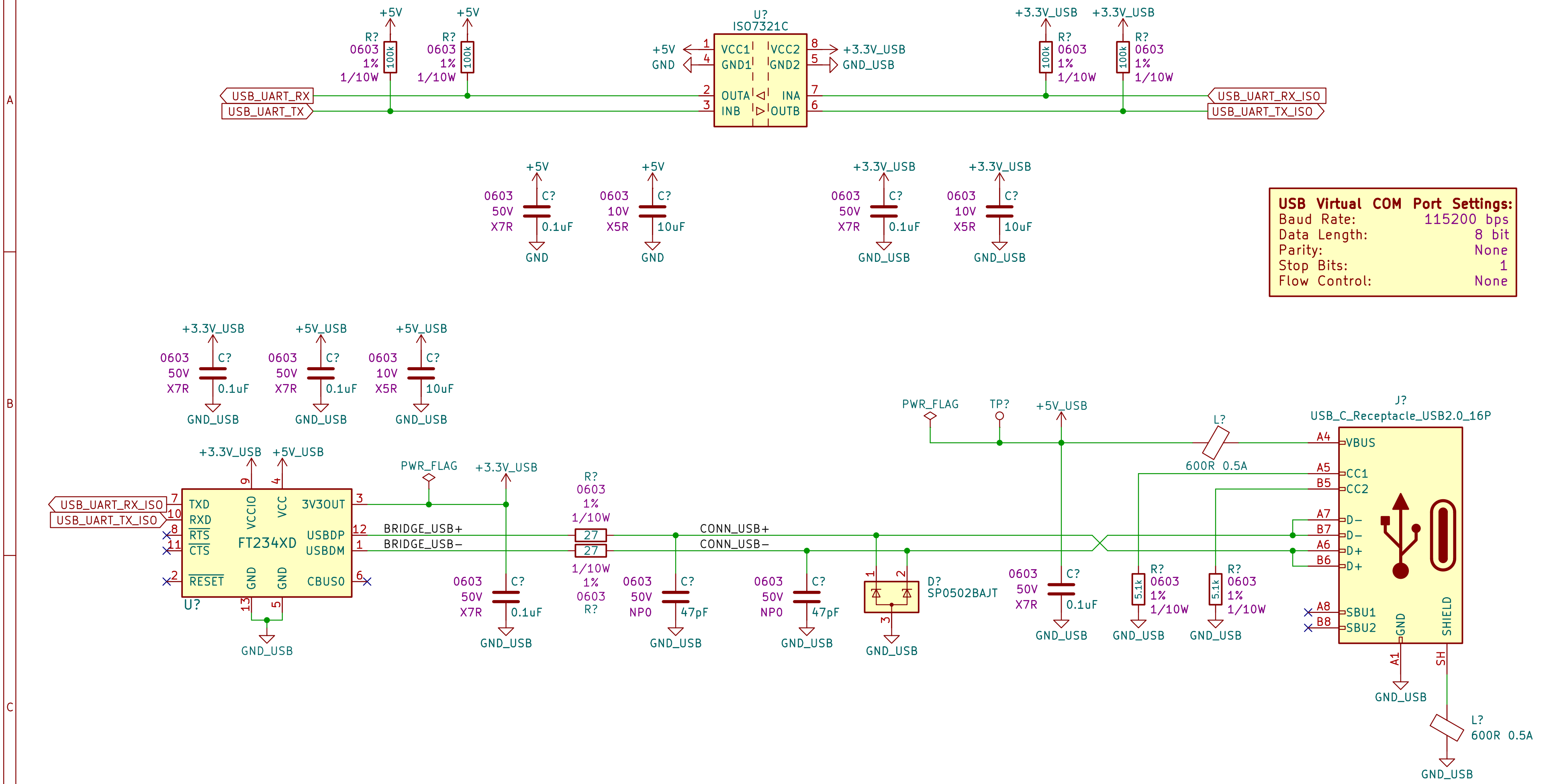
Id: 5/7



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Sheet: /USB UART Bridge/

File: USB_UART_Bridge.kicad_sch

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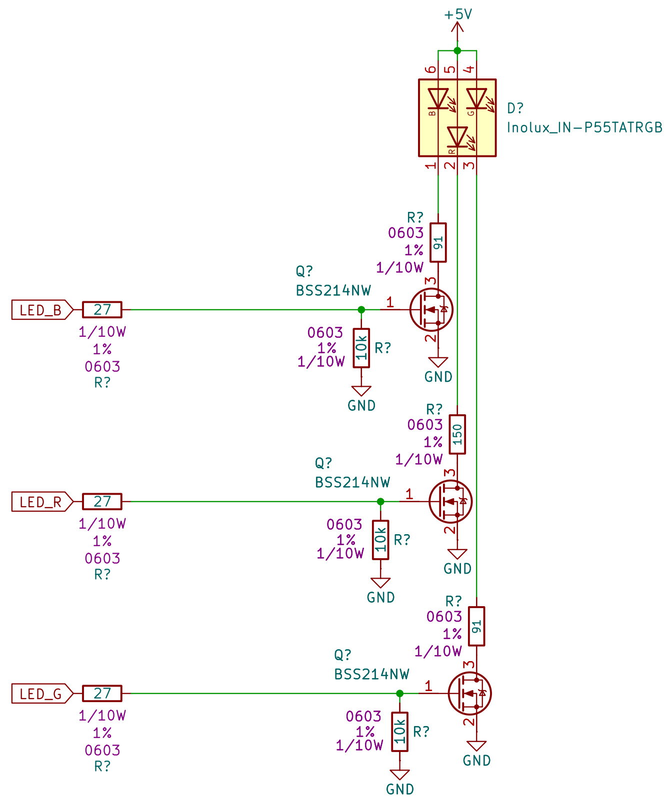

Based on the datasheet, the typical forward current (I_F) for testing is 20mA,
with an absolute maximum of 25mA.
For "maximum brightness" while maintaining reliability, I recommend targeting 20mA.

Using Ohm's Law (R = (VCC − VF)*IF with VCC = 5V:

Channel	Typ Vf	Calculation (20mA)	Recommended Res.	Max Bright (25mA)
Red	2.0V	(5 − 2.0) / 0.020	150 Ohm	120 Ohm
Green	3.2V	(5 − 3.2) / 0.020	91 Ohm (Standard)	72–75 Ohm
Blue	3.2V	(5 − 3.2) / 0.020	91 Ohm (Standard)	72–75 Ohm

Design Notes:
- Power Dissipation: At 20mA, the resistors dissipate approx 36–60mW. Standard 0603 or 0805 SMD resistors are more than sufficient.
- Safety: I strongly recommend the 20mA values. Pushing to 25mA provides a marginal increase in brightness but reduces the LED's lifespan and increases heat.

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Sheet: /RGB LED/

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